

Customer Shopping Behavior Analysis

Overall Project Outcome Report

An end-to-end data analytics project covering data cleaning, exploratory analysis, SQL-based business questions, visualization, and dashboarding on 3,900 retail transactions.

Dataset Size	3,900 transactions × 18 columns
Tools Used	Python (pandas, matplotlib), PostgreSQL, Power BI
Report Sections	Overview, Methodology, Key Findings, Visuals, Recommendations

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across four product categories. The objective was to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior in order to guide strategic business decisions.

The workflow followed an industry-standard analytics pipeline: data preparation and exploratory analysis in Python, structured business-question analysis in SQL (PostgreSQL), visual analytics in both Python (matplotlib) and Power BI, and a final report with actionable recommendations.

Dataset Summary

Metric	Value
Rows	3,900
Columns	18
Missing Data	37 values in Review Rating (imputed via category median)
Key Feature Groups	Demographics · Purchase Details · Shopping Behavior

2. Methodology

Python — Data Preparation & EDA

- Loaded and inspected the dataset with pandas (`df.info()`, `df.describe()`)
- Imputed 37 missing Review Rating values using the median rating per product category
- Standardized column names to snake_case
- Engineered age_group (quartile-based) and purchase_frequency_days features
- Dropped the redundant Promo Code Used column after a consistency check against Discount Applied
- Loaded the cleaned DataFrame into PostgreSQL for structured querying

SQL — Business Question Analysis

Ten structured business questions were answered directly in PostgreSQL, covering revenue drivers, discount behavior, customer segmentation, shipping preferences, and product performance. Full query set is included in the project repository (`customer_behavior_sql_queries.sql`).

Visualization — Python & Power BI

Each SQL finding was visualized twice: once with matplotlib for the analytical notebook, and once as an interactive Power BI dashboard with slicers for subscription status, gender, category, and shipping type.

3. Key Findings

#	Question	Finding
1	Revenue by gender	Male customers generated \$157,890 vs. \$75,191 from female customers
2	High-spend discount users	839 customers used a discount yet still spent above the average purchase amount
3	Top-rated products	Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)
4	Shipping type spend	Express avg. \$60.48 vs. Standard avg. \$58.46 — a modest premium-shipping effect
5	Subscribers vs. non	Non-subscribers drive far more total revenue (\$170,436 vs. \$62,645) despite similar avg. spend (~\$59-60)
6	Discount-dependent products	Hat (50%), Sneakers (49.7%), Coat (49.1%), Sweater (48.2%), Pants (47.4%) of purchases involved a discount
7	Customer segments	Loyal: 3,116 · Returning: 701 · New: 83 — the base is overwhelmingly loyal
8	Top products per category	Jewelry, Blouse, Sandals, and Jacket lead their respective categories
9	Repeat buyers & subscription	Of repeat buyers (>5 purchases), 2,518 are non-subscribers vs. 958 subscribers
10	Revenue by age group	Young Adult (\$62,143) leads, followed by Middle-aged, Adult, and Senior — a fairly even spread

Headline Takeaways

- **Revenue is subscriber-agnostic:** non-subscribers contribute ~73% of total revenue purely on volume, meaning the subscriber program is not yet a primary revenue lever.
- **Customer base is loyalty-heavy:** 80% of customers already qualify as "Loyal" (>10 previous purchases), suggesting strong retention but a thin pipeline of genuinely new customers (just 83).
- **Discounting is broad, not targeted:** the top 5 discount-dependent products cluster tightly around 47-50%, indicating discounting is used as a general sales lever rather than for specific underperforming items.
- **Gender revenue gap is substantial:** male customers generate roughly double the revenue of female customers, warranting further investigation into category mix or basket size by gender.

4. Visual Analysis

The following charts were generated with matplotlib directly from the cleaned dataset, cross-validated against the SQL query results above.

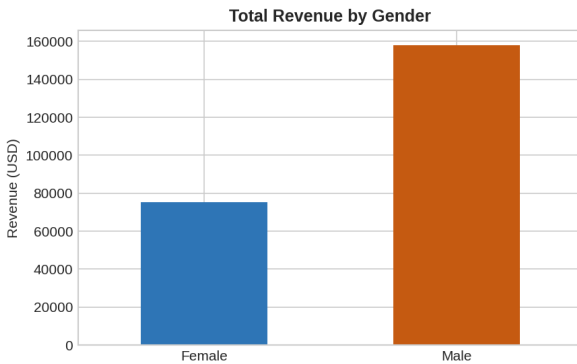


Figure 1. Total revenue by gender

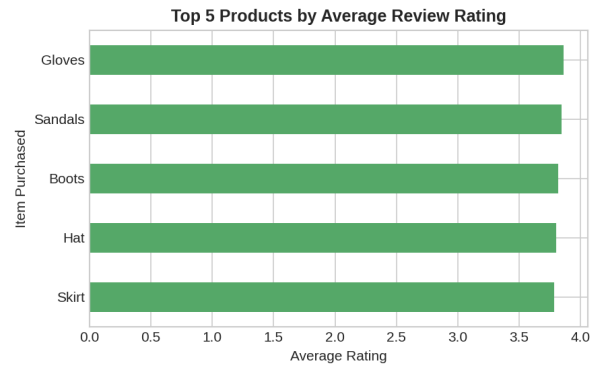


Figure 2. Top 5 products by average review rating



Figure 3. Average purchase amount — Standard vs. Express shipping

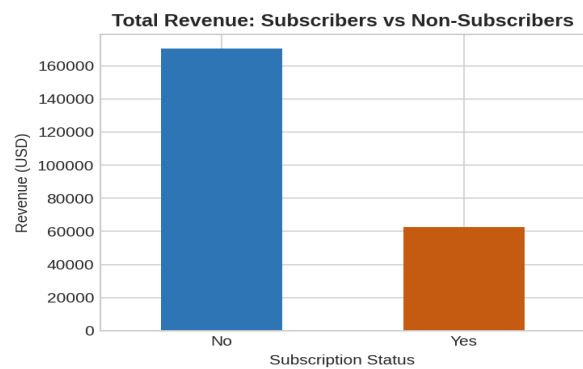


Figure 4. Total revenue — subscribers vs. non-subscribers

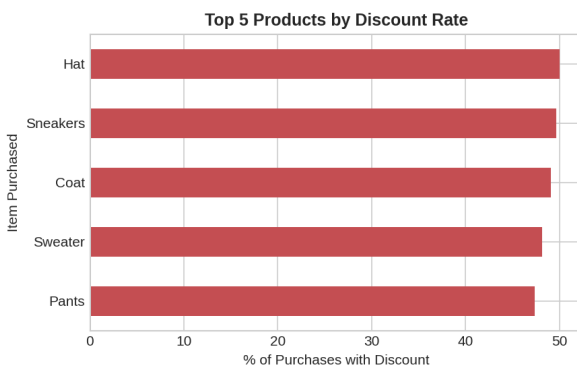


Figure 5. Top 5 products by discount rate



Figure 6. Customer segmentation (New / Returning / Loyal)

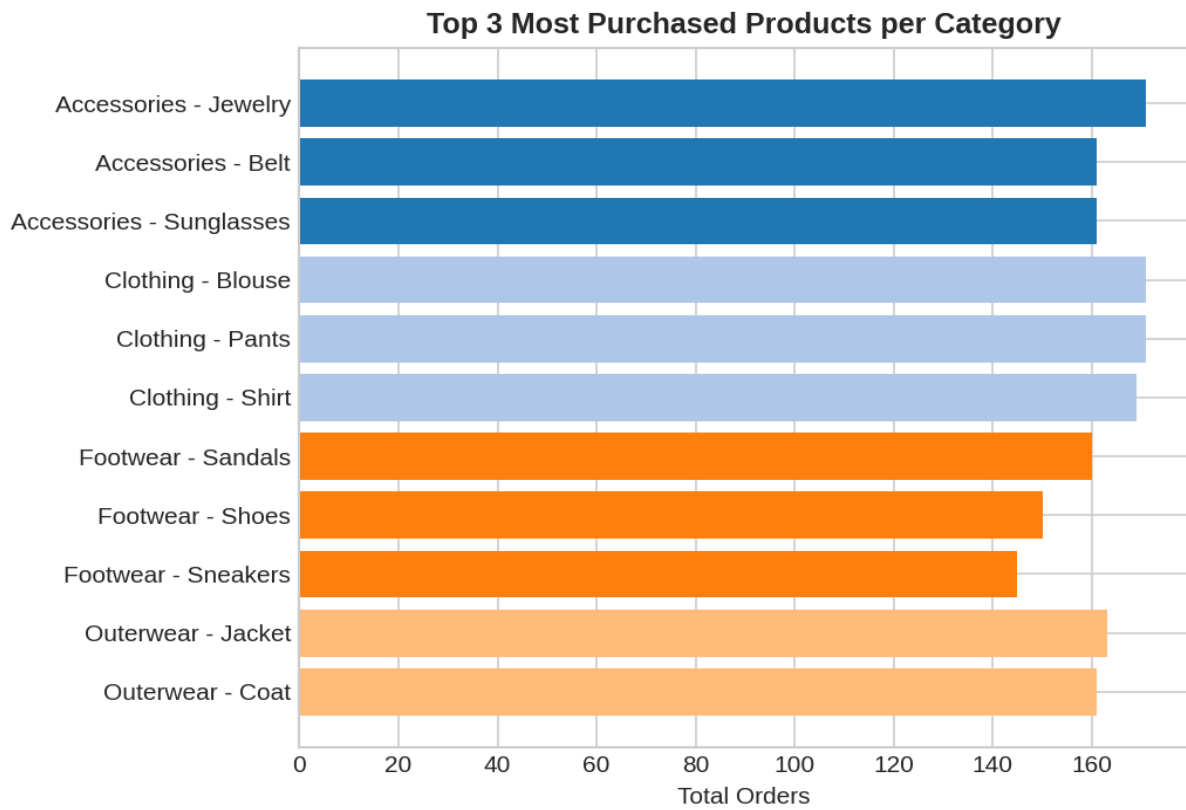


Figure 7. Top 3 most purchased products per category

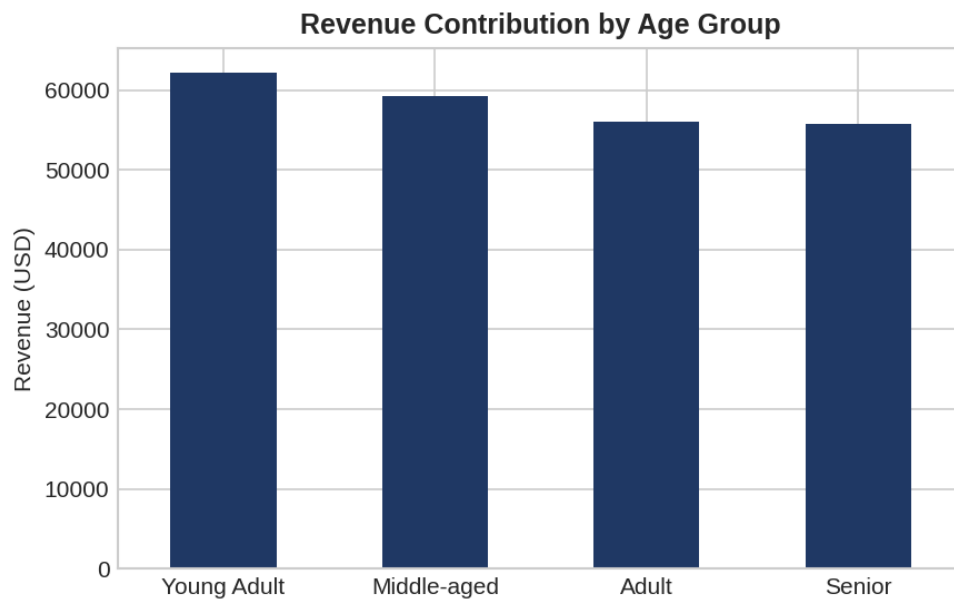


Figure 8. Revenue contribution by age group

5. Power BI Dashboard

An interactive Power BI dashboard was built to present these insights to stakeholders, with filters for subscription status, gender, category, and shipping type. Key headline metrics on the dashboard:

3.9K	\$59.76	3.75
Number of Customers	Average Purchase Amount	Average Review Rating

The dashboard additionally breaks down revenue and sales by category and by age group, and visualizes the subscriber split (27% subscribed vs. 73% not), letting stakeholders filter interactively rather than reading static tables.

6. Business Recommendations

Recommendation	Rationale
Boost Subscriptions	Promote exclusive benefits for subscribers — subscribers currently account for only 27% of the base and a minority of total revenue
Customer Loyalty Programs	Reward repeat buyers to move Returning customers into the Loyal segment and sustain the already-strong loyalty base
Review Discount Policy	Balance the broad ~47-50% discount rate on top products against margin control — consider more targeted discounting
Product Positioning	Highlight top-rated (Gloves, Sandals, Boots) and best-selling (Jewelry, Blouse, Sandals, Jacket) products in campaigns
Targeted Marketing	Focus acquisition and retention efforts on high-revenue age groups (Young Adult) and express-shipping users

7. Conclusion

This project demonstrates a complete, professional data analytics workflow — from raw transactional data to cleaned datasets, structured SQL analysis, Python and Power BI visualizations, and stakeholder-ready recommendations. The customer base is loyal and revenue is broadly distributed across age groups, but there is clear opportunity to grow subscription participation, tighten discount targeting, and capitalize on top-performing products and demographics.

Full project files (notebook, SQL scripts, dashboard, dataset) are available in the accompanying repository.